

Wollo University
Department of Health Informatics
Internet Programming I

Instructor Name:- Daniel Bekele

Chapter 3: Web Design and Development

3.1. Mark-up languages

A markup language uses **text-based tags** to define the structure and presentation of a document or data, allowing computers to understand and process its content. Popular examples include **HTML** (Hyper Text Markup Language) for web pages and **XML** (Extensible Markup Language) for data storage and exchange, which use tags like <h1> or <p> to label and format text.

How Markup Languages Work

- **Tags:** These are special keywords, often enclosed in angle brackets (e.g., <tag>), that provide instructions to the computer.
- **Elements:** Tags define different parts of a document, such as headings, paragraphs, or images, and can be nested to show relationships between them.
- **Purpose:** By using tags, markup languages control how content is displayed, its structure, and its meaning for both humans and machines.

Examples of Markup Languages

- **HTML (Hyper Text Markup Language):** Used to create web pages by specifying the layout, formatting, and hyperlinks of documents.
- **XML (Extensible Markup Language):** Designed for storing and transporting data, allowing users to create their own custom tags to describe the data's meaning.
- **XHTML:** A stricter version of HTML, adhering to XML's rules for enhanced validation and compatibility.

Why Use Markup Languages?

- **Structure and Presentation:** They provide a clear way to define the visual layout and logical structure of digital content.

- **Data Exchange:** Languages like XML enable easy and standardized sharing of data between different software applications and platforms.
- **Accessibility:** Proper use of markup helps ensure that content is understandable by assistive technologies, making websites accessible to people with disabilities.

3.1.1. HTML

HTML (Hyper Text Markup Language) is **the most basic building block of the Web**. It defines the meaning and structure of web content.

Example

```
<!DOCTYPE html>
<html>
<head>
  <title>My Web Page</title>
</head>
<body>
  <h1>Welcome!</h1>
</body>
</html>
```

The `<html>` tag: This tag is the root element of an HTML page. All other content, including the `<head>` and `<body>` sections, resides within the `<html>` tag.

The `<!DOCTYPE html>` declaration at the beginning of an HTML document tells a web browser to render the page in **standards mode**, using the **HTML5** specification, rather than **quirks mode**. It is not an HTML tag itself but an instruction that ensures the webpage is displayed correctly according to the latest web standards and technologies.

Key Aspects

- **Required First Line:** All HTML documents must start with a doctype declaration, and it must appear before any other HTML tags.
- **HTML5 Standard:** `<!doctype html>` specifically declares the document to be using the HTML5 standard.
- **Standards Mode:** Its presence triggers "standards mode," where the browser interprets the page according to the defined standards.

- **Correct Rendering:** This ensures that the HTML and CSS are rendered correctly by the browser, enabling the use of modern web technologies.
- **No Closing Tag:** There is no closing tag for the doctype declaration.

3.1.2. XML

XML (Extensible Markup Language) is a universal markup language used for storing, transporting, and reconstructing data in a way that is both human-readable and machine-readable. It defines rules for creating custom, user-defined tags that describe data, enabling it to represent complex, hierarchical data relationships across different systems and applications.

Key Characteristics

- **Extensible:** Users define their own tags tailored to the specific data they are representing, unlike HTML which uses predefined tags.
- **Markup Language:** It provides a set of rules for encoding documents, essentially adding metadata to data.
- **Self-Descriptive:** Tags provide a clear, understandable description of the data they contain.
- **Hierarchical Structure:** Data is organized in a tree-like structure, allowing for complex relationships to be represented.
- **Standardized:** The underlying syntax is a **W3C standard** (W3C standard is a technical guideline or specification developed by the World Wide Web Consortium to ensure the interoperability, accessibility, security, and consistency of the World Wide Web.), ensuring that XML data can be reliably parsed and shared across different platforms and applications

How XML Works

1. **Tags:** XML uses tags, enclosed in angle brackets (<tag>), to define elements.
2. **Elements:** An XML element consists of a start tag, an end tag, and the data in between.
Example: <book><title>The Great Alexander </title></book>
3. **Structure:** Elements are nested within each other to create a hierarchical structure.
4. **Root Element:** Every XML document must have a single root element that contains all other elements.

5. **Attributes:** Elements can have attributes, which provide additional information about the element.

Example: `<book category="fiction">...</book>`

Common Use Cases

- **Data Exchange:** Facilitates information exchange between different systems, such as web services and databases.
- **Configuration Files:** Used to store and configure application settings.
- **Document Storage:** Provides a flexible format for storing documents.
- **Web Applications:** As seen in Android development, XML is used to define user interfaces and other app resource

3.2. Web Scripting Languages

Web scripting languages are programming languages used to create dynamic and interactive web content. They can be broadly categorized into client-side and server-side scripting languages.

Client-Side Scripting Languages:

These languages execute within the user's web browser, directly interacting with the HTML and CSS of a webpage to enhance user experience.

- **JavaScript:** The most prevalent client-side scripting language, enabling interactive elements, animations, form validation, and dynamic content manipulation.

Server-Side Scripting Languages:

These languages execute on the web server, handling tasks like database interactions, user authentication, and generating dynamic HTML content before sending it to the client's browser.

- **PHP:** A widely used server-side language, particularly popular for content management systems like WordPress.
- **Python:** A versatile language used for web development frameworks like Django and Flask, known for its readability and extensive libraries.
- **Ruby:** An object-oriented language often used with the Ruby on Rails framework, emphasizing developer productivity.

- **Node.js:** A JavaScript runtime environment that allows JavaScript to be used for server-side development, enabling full-stack JavaScript applications.
- **ASP.NET:** A framework developed by Microsoft for building web applications and services, often using C#.
- **Perl:** A powerful language known for its text processing capabilities and used in various web development contexts.

3.2.1. ASP

ASP stands for **Active Server Pages**, a server-side scripting technology developed by Microsoft. It is used to create **dynamic and interactive web pages** that can interact with databases and respond to user input. ASP allows developers to embed scripts (usually written in VBScript or JavaScript) within HTML pages to perform server-side processing before sending the final page to the user's browser.

Key Features of ASP:

1. **Server-Side Execution** – ASP code runs on the web server, not on the user's computer, ensuring secure and efficient processing.
2. **Dynamic Content** – Pages can display changing data (e.g., from a database or user form).
3. **Database Connectivity** – ASP can connect to databases like SQL Server or Access using ADO (ActiveX Data Objects).
4. **Reusable Components** – Developers can use reusable objects and components to simplify coding.
5. **Platform Integration** – Works well with other Microsoft technologies such as IIS (Internet Information Services).

3.2.2. CGI

The Common Gateway Interface (CGI) is a standard protocol in web technology that enables web servers to execute external programs or scripts in response to user requests. It acts as an interface between the web server and these external programs, facilitating the generation of dynamic web content.

How CGI Works:

- **Client Request:** A user's browser sends an HTTP request to the web server, often involving a form submission or a request for dynamic content.
- **Server Processing:** If the request is for a dynamic resource, the web server, recognizing a CGI script (often located in a cgi-bin directory), prepares to execute it.
- **Invoking the CGI Script:** The web server launches the specified CGI script as a separate process. It passes relevant information from the HTTP request to the script through environment variables and standard input.
- **Script Execution:** The CGI script processes the request, potentially interacting with databases or other resources, and generates dynamic content, typically in HTML format.
- **Output and Response:** The script sends its generated content to standard output, which the web server captures. The server then packages this output as an HTTP response and sends it back to the user's browser.

Key Features and Significance:

- **Dynamic Content Generation:** CGI was instrumental in moving the web beyond static HTML pages by enabling the creation of interactive and personalized content.
- **Language Independence:** CGI scripts can be written in various programming languages (e.g., Perl, Python, C, Shell scripts), offering flexibility to developers.
- **Historical Impact:** CGI played a crucial role in the early development of the World Wide Web, laying the foundation for modern server-side programming.

Limitations:

- **Performance Overhead:** Each CGI request typically involves launching a new process, leading to performance overhead, especially on high-traffic websites.
- **Scalability Challenges:**
- The process-per-request model can limit scalability and efficiency compared to more modern server-side technologies.

3.2.3. PHP

It was abbreviated previously as Personal Home Page. It is a programming language widely used to build web applications or websites. It is the server-side scripting language encoded with HTML

to develop Dynamic website, Static website or Web application. Now **PHP** stands for **Hypertext Preprocessor**, and it is a widely used **server-side scripting language** designed for creating dynamic and interactive web pages. PHP code is executed on the server, and the result is sent to the user's browser as plain HTML.

Key Features of PHP:

1. **Server-Side Processing** – PHP runs on the web server, allowing it to handle tasks like form submission, file handling, and database access before sending the page to the browser.
2. **Database Integration** – PHP easily connects to databases such as MySQL, PostgreSQL, and Oracle, making it ideal for data-driven websites.
3. **Cross-Platform Compatibility** – It works on different operating systems, including Windows, Linux, and macOS.
4. **Open Source** – PHP is free to use and supported by a large developer community.
5. **Embedded in HTML** – PHP can be written directly inside HTML code, making it easy to integrate with web page design.
6. **Extensive Library Support** – It includes many built-in functions and can use external libraries for additional functionality.

Common Uses of PHP:

- Developing dynamic websites and web applications.
- Building content management systems (CMS) like WordPress, Joomla, and Drupal.
- Handling form data and user authentication.
- Generating dynamic page content based on database data.

PHP code is typically embedded within HTML and is executed on a server. Here are some simple PHP code examples:

This is a fundamental example demonstrating how to output text to the browser.

```
<!DOCTYPE html>
<html>
<head>
  <title>PHP Test</title>
</head>
```

```

<body>
  <?php
    echo "Hello World!";
  ?>
</body>
</html>

```

Explanation:

- <?php ... ?>: These are the PHP opening and closing tags. Any code within these tags will be interpreted by the PHP engine.
- echo: This is a PHP statement used to output strings or values.

Variables and Basic Operations:

This example demonstrates declaring variables, assigning values, and performing a simple arithmetic operation.

```

<?php
  $name = "Ali";
  $age = 30;
  $year_born = date("Y") - $age; // Calculates year born based on current year
  echo "My name is " . $name . " and I am " . $age . " years old. I was likely born in " .
  $year_born . ".";
?>

```

Explanation:

- \$name, \$age, \$year_born: These are variables, denoted by the \$ prefix.
- date("Y"): This is a built-in PHP function that returns the current year.
- .: The dot operator is used for string concatenation (joining strings together).

3.2.4. ColdFusion

ColdFusion" can refer to a web development platform and scripting language created by Adobe (also known as CFML). The programming platform facilitates rapid web development, allowing developers to create dynamic web pages and applications that interact with databases and other

back-end systems. CFML is a server-side scripting language that uses HTML-like tags, making it accessible to HTML developers.

ColdFusion as a Web Development Platform

- **Purpose:** To enable rapid development of web applications and dynamic websites.
- **Components:** It consists of the ColdFusion Markup Language (CFML) and the ColdFusion application server.
- **CFML:** This is the scripting language used to build applications, which runs on various platforms, including Java and .NET.

Key Features:

- **Ease of Use:** CFML's tag-based syntax was designed to be familiar to HTML developers.
- **Database Integration:** It simplifies connecting web applications to databases.
- **Productivity:** The platform is known for out-of-the-box capabilities that boost development speed and efficiency.

Implementations:

Besides Adobe's commercial version, other implementations of the CFML engine exist, including the open-source Lucee and Railo.

3.3. Java

Java is a popular, high-level, object-oriented programming language and computing platform used to develop a wide range of applications, from mobile and web apps to enterprise software, big data, and Internet of Things (IoT) devices. Key features include its "write once, run anywhere" capability, which allows compiled Java code (bytecode) to run on any platform with a Java Virtual Machine (JVM), providing high portability and platform independence.

Key Characteristics and Concepts

- **Object-Oriented:** Java models real-world entities as objects created from classes, which serve as blueprints for these objects.

- **Platform-Independent:** Java achieves platform independence through the Java Virtual Machine (JVM). Code is compiled into Java bytecode, which can then run on any system with a compatible JVM.
- **"Write Once, Run Anywhere":** This principle ensures that Java code can execute on any platform without needing to be recompiled for that specific system.
- **Memory-Safe:** Java is designed to be memory-safe, preventing common memory-related errors.
- **Large Standard Library:** Java includes an extensive Application Programming Interface (API) that provides pre-written code for tasks like networking, security, and web services.

How Java Works

1. **Development:** Developers write Java code using the Java programming language.
2. **Compilation:** The Java code is compiled into intermediate bytecode.
3. **Execution:** The Java Virtual Machine (JVM) interprets and runs this bytecode on the specific hardware and operating system.
4. **Java Platform:** This platform consists of the JVM, the Java API, and a complete development environment, providing a reliable foundation for applications.

Common Uses of Java

Java is used extensively in various areas of software development:

- **Mobile Applications:** Powering many Android apps.
- **Web Applications:** Creating server-side technologies and web services.
- **Enterprise Software:** Building large-scale business applications.
- **Big Data:** Developing systems for processing large datasets.
- **Internet of Things (IoT):** Creating software for connected devices.

A simple Java program typically consists of a class containing a main method, which serves as the entry point for program execution. Within the main method, statements can be written to perform various actions, such as printing output to the console.

Here is a simple "Hello, World!" program in Java:

```
public class HelloWorld {  
    public static void main(String[] args) {  
        System.out.println("Hello, World!");  
    }  
}
```

Explanation:

- **public class HelloWorld:**

This line declares a public class named HelloWorld. In Java, all code resides within classes. The class name should match the filename (e.g., HelloWorld.java).

- **public static void main(String[] args):** This is the main method, the starting point of the program.
 - **public:** Makes the method accessible from outside the class.
 - **static:** Allows the main method to be called without creating an object of the class.
 - **void:** Indicates that the method does not return any value.
 - **String[] args:** Represents an array of strings that can be used to pass command-line arguments to the program.
- **System.out.println("Hello, World!");:** This statement prints the string "Hello, World!" to the console, followed by a new line. System.out refers to the standard output stream, and println() is a method that prints the given argument.

3.3.1. Servlets & Applets

Servlets and Applets are both Java technologies, but they serve distinct purposes and operate in different environments within a web application architecture.

Servlet:

- **Server-side execution:** A Servlet is a Java program that runs on a web server (within a Servlet container like Apache Tomcat).
- **Web application backbone:** Servlets are primarily used to handle HTTP requests from clients, process data, interact with databases, and generate dynamic web content (like HTML pages) to be sent back to the client's browser.

- **No graphical user interface:** Servlets typically do not have a direct graphical user interface. Their output is usually in the form of HTML, XML, JSON (JavaScript Object Notation), or other data formats consumed by the client.
- **Persistence:** Once a servlet is loaded, it remains in memory and can handle multiple requests, making it efficient for web applications.

Applet:

- **Client-side execution:** An Applet is a small Java program designed to be embedded within an HTML page and downloaded to a client's web browser for execution.
- **Interactive user interface:** Applets can provide rich and interactive graphical user interfaces directly within the browser, enhancing the user experience beyond what static HTML and basic JavaScript could offer.
- **Limited access to client resources:** Due to security restrictions, applets have limited access to the client machine's local resources.
- **Declining usage:** While historically significant, applet usage has declined significantly due to security concerns, browser compatibility issues, and the rise of alternative client-side technologies like JavaScript frameworks.

Key Differences Summarized:

Feature	Servlet	Applet
Location	Server-side	Client-side (embedded in web page)
Purpose	Dynamic web content generation, data processing	Interactive user interfaces in the browser
Execution	In a Servlet container on a web server	In a Java-enabled web browser
GUI	None (generates content for client GUI)	Has a graphical user interface
Persistence	Persistent (handles multiple requests)	Downloaded and executed for each page load

3.3.2. JSP

JSP, or Jakarta Server Pages, JSP; formerly Java Server Pages is a Java-based web technology that allows developers to create dynamic web pages by embedding Java code within HTML, XML, or other document types. It was originally developed by Sun Microsystems as part of the Java 2

Enterprise Edition (J2EE) and serves as an alternative to technologies like PHP and ASP. JSPs separate content presentation from dynamic logic, using a server-side Java execution model to generate content that is then sent to the user's web browser.

Key Features and Concepts

- **Dynamic Content Generation:** JSPs provide a way to insert dynamic information into web pages by embedding Java code directly into the static template data (like HTML).
- **Server-Side Execution:** The Java code within a JSP is processed on the web server by a web container, which translates the JSP into a servlet, compiles it, and then executes the servlet to produce the dynamic content.
- **Separation of Concerns:** JSPs enable the separation of the presentation layer (HTML/XML) from the business logic (Java code), catering to different developers (HTML designers and Java developers).
- **JSP Elements:** A JSP page consists of static template data and JSP elements, such as scriptlets, expressions, and directives, that define the dynamic behavior.
- **JavaBeans and Custom Tags:** The technology supports the use of JavaBean components and custom tags to provide reusable functionality and simplify the development of dynamic web pages.
- **Part of Java Enterprise:** JSP is a key component of the Java Enterprise Edition (formerly J2EE), which provides a standardized platform for developing enterprise-level web applications.

How JSPs Work

1. **Request:** A user's browser requests a JSP file.
2. **Translation:** The web container translates the JSP file into a Java Servlet code.
3. **Compilation:** The Servlet code is then compiled into a Java bytecode class file.
4. **Execution:** The web container loads the compiled servlet and executes it, processing any dynamic Java code.
5. **Response:** The servlet generates dynamic content (typically HTML), which is sent back to the user's browser as the response.

3.4. ActiveX

ActiveX is a defunct/obsolete Microsoft software framework based on its Component Object Model (COM) and Object Linking and Embedding (OLE) technologies, designed to run small applications called ActiveX controls on the internet and within Windows applications. While it provided powerful features for creating interactive web content and combining applications, ActiveX controls posed significant security risks because they could access local files and system settings, leading to their deprecation by Microsoft.

How ActiveX Works

- **Building Blocks:** ActiveX controls are essentially small programs that act as building blocks for larger applications.
- **Web Browsers and Applications:** They were designed to extend the functionality of web pages and other Windows applications, enabling features like animated content, data collection, and custom user interfaces.
- **Component Reusability:** By leveraging COM and OLE, ActiveX allowed for the reuse of code and the linking of different programs together.

Security Concerns

- **Unrestricted Access:** ActiveX controls often had unrestricted access to a user's computer, meaning they could read local files, modify the registry, and even control the PC remotely.
- **Potential for Abuse:** This unrestricted access made them vulnerable to malicious use, as they could be used to install unwanted software, damage data, or steal information without the user's explicit agreement.

Current Status

- **Deprecated Technology:** Microsoft has largely deprecated ActiveX technology due to its inherent security risks.
- **Security Recommendations:** Microsoft recommends keeping ActiveX controls disabled by default and only enabling them when absolutely necessary for specific trusted applications or websites.

Why It Matters

- **Internet Explorer:** ActiveX was most prominently used with Internet Explorer to provide interactive features not natively supported by web browsers at the time.
- **Security Practices:** Understanding ActiveX is important for recognizing its security implications, as it represents a significant shift in how web applications interacted with user systems.

3.5. Editors/Development Environments

Editors and Integrated Development Environments (IDEs) for web development offer various features to aid in coding, debugging, and project management. The choice often depends on individual preferences, project requirements, and the specific technologies being used.

Popular Editors and IDEs for Web Development:

- **Visual Studio Code (VS Code):** A highly popular and versatile open-source code editor developed by Microsoft. It offers extensive features for web development, including intelligent code completion (IntelliSense), debugging tools, integrated Git control, and a vast marketplace of extensions for various languages and frameworks (HTML, CSS, JavaScript, React, Angular, Vue.js, Node.js, Python, PHP, etc.).
- **Sublime Text:** A lightweight, fast, and feature-rich text editor known for its speed and efficiency. It is highly customizable and offers powerful features like multiple selections, command palette, and a robust plugin ecosystem.
- **WebStorm:** A commercial IDE developed by JetBrains, specifically designed for web development. It provides advanced features for JavaScript, HTML, CSS, and related technologies, including intelligent code analysis, refactoring tools, and integrated debugging.
- **Atom:** An open-source, hackable text editor developed by GitHub. It is highly customizable and offers a wide range of packages and themes to extend its functionality.
- **Brackets:** An open-source editor from Adobe, focused on front-end web development. It features live preview, inline editors for CSS and JavaScript, and preprocessor support.
- **Vim/Neovim & Emacs:** Highly configurable and powerful text editors favored by many experienced developers for their efficiency and customization options, though they have a steeper learning curve.

Integrated Development Environments (IDEs) with Web Development Support:

- **IntelliJ IDEA Ultimate:** A comprehensive IDE from JetBrains with extensive support for various programming languages and frameworks, including robust web development features.
- **PyCharm (for Python web development):** Another JetBrains IDE, tailored for Python development but also offering strong support for web technologies when used with frameworks like Django or Flask.
- **PhpStorm (for PHP web development):** A JetBrains IDE specifically for PHP development, including features for front-end web technologies.
- **Eclipse:** A long-standing open-source IDE with various plugins and distributions for different development needs, including web development.
- **NetBeans:** Another open-source IDE that supports multiple languages and offers features for web development.

Key Features to Consider:

- **Code Completion and IntelliSense:** Suggestions and auto-completion for faster and more accurate coding.
- **Debugging Tools:** Features for identifying and fixing errors in code.
- **Version Control Integration:** Seamless integration with systems like Git for managing code changes.
- **Extensions/Plugins:** Ability to extend functionality with additional tools and language support.
- **Customization:** Options for personalizing the interface, keybindings, and workflows.
- **Performance:** Speed and responsiveness of the editor/IDE.
- **Language Support:** Compatibility with the programming languages and frameworks used in the project.

3.5.1. Notepad/Wordpad Editor

While standard Notepad and WordPad are insufficient for serious web development, **Notepad++** is a highly recommended free, open-source Windows text editor with features like syntax highlighting for HTML, CSS, and PHP, and code folding to make coding efficient. Other powerful,

though more feature-rich, options include dedicated code editors like Visual Studio Code and Sublime Text, and online tools like [Hyper Notepad](#).

Why Notepad and WordPad aren't Ideal

- **Limited Functionality:** Standard Notepad and WordPad lack essential features for web development, such as syntax highlighting, code folding, and advanced find/replace functions.
- **Plain Text Focus:** They are primarily for plain text editing, not for the complex syntax and structure of web development languages like HTML, CSS, or JavaScript.

3.5.2. Sublime editor

Sublime Text is a fast, proprietary source code editor for web development that offers a minimal interface, syntax highlighting, code folding, and extensive customization through plugins. It supports multiple operating systems, can be extended with packages from a built-in Package Control system, and provides features like code completion and integrated terminals for efficient coding in languages such as HTML, CSS, and JavaScript.

Key Features for Web Development

- **Speed and Performance:** Written in C++ and Python, Sublime Text is known for its exceptional speed and lightweight nature, allowing for a smooth editing experience.
- **Syntax Highlighting:** The editor automatically highlights code elements in different colors, making it easier to read and understand complex web code.
- **Code Folding:** This feature allows developers to collapse sections of code, simplifying navigation through large files and focusing on specific parts of the project.
- **Package Control System:** A built-in system allows users to easily install and manage community-developed plugins, which expand the editor's functionality with new features and support for more languages.
- **Cross-Platform Compatibility:** Sublime Text is available for Windows, macOS, and Linux, making it accessible to developers on various operating systems.
- **Customization:** Developers can highly personalize their coding environment by customizing themes and installing extensions to suit their workflow and project needs.

How it Works for Web Development

1. **Install the Editor:** Download and install Sublime Text on your computer.
2. **Open Your Project:** Open the folder containing your website's files to gain access to all the project files in the sidebar.
3. **Write and Edit Code:** Use the editor to write and edit HTML, CSS, JavaScript, and other web development languages.
4. **Install Plugins:** Use Package Control to install plugins that add features like code auto completion, advanced search functions, or specialized tools for front-end development.
5. **Automate Tasks:** Utilize built-in features and plugins to automate repetitive coding tasks, allowing you to focus on the core aspects of development.

3.5.3. Freeware/Shareware

Freeware is software available completely free of cost and without a trial period, while **Shareware** is proprietary software (**Proprietary software** is a type of software owned by a developer or company that holds exclusive rights to its source code, which is kept private.) offered for a limited trial or with restricted features, requiring payment for full or continued use. For web development, this means tools like free web browsers or editors (freeware) are available without cost, while trial versions of more complex software may be offered for evaluation before purchase (shareware).

Freeware

- **Definition:** Completely free to use, with no payment required.
- **Features:** Users receive the fully functional software without restrictions.
- **Cost:** No purchase or subscription is necessary.
- **Examples in Web Development:**
 - **Web Browsers:** Google Chrome, Mozilla Firefox, Microsoft Edge are examples of freeware browsers used to test websites.
 - **Code Editors:** Some advanced code editors are available as freeware, offering powerful tools for free.
 - **Web Servers:** Tools like the Apache HTTP Server are free and open-source, which is a subset of freeware where source code is also available.

Shareware

- **Definition:** Proprietary software offered for trial use, requiring payment for full access or continued use after the trial period ends.
- **Features:** Often comes with a limited feature set, a time-limited trial, or displays reminders to pay for the software.
- **Cost:** Users are expected to pay a registration fee to the developer to continue using the software after the trial.
- **Examples in Web Development:**
 - **Advanced Code Editors:** Some professional web development code editors may be offered as shareware, allowing users to try out advanced features before buying.
 - **Design Software:** GNU Image Manipulation Program (GIMP), GIMP is a cross-platform image editor available for GNU/Linux, macOS, Windows and more operating systems. It is free software, you can change its source code and distribute your changes. GIMP is an example of a free and open-source photo editor for web design, while other graphic design tools might be offered as shareware with a limited trial.

Modern Terms:

The terms "trialware" or "freemium" are now more common than "shareware".

- **Trialware:** Software available for a limited time to try out.
- **Freemium:** Software that offers a free version with limited features and a premium version with full functionality for a fee.

Key Differences

- **Cost:** Freeware is always free, while shareware requires payment for continued or full use.
- **Access:** Freeware offers the full, functional product upfront. Shareware provides a limited version or access for a specific period only.

3.5.4. MS FrontPage

MS FrontPage was a discontinued What You See Is What You Get (WYSIWYG) HTML editor and website administration tool from Microsoft that allowed users to create and manage websites without extensive knowledge of HTML or programming. It was part of the Microsoft Office suite and featured a user-friendly interface with features like drag-and-drop page layout,

templates, and integration with other Office applications. FrontPage is no longer available and has been replaced by other tools, but it was a significant tool for web development in the late 1990s and early 2000s.

Key Features and Functionality

- **WYSIWYG Editor:** Users could design web pages visually, much like using a word processor, with the software automatically generating the necessary HTML code.
- **Website Administration:** FrontPage included tools for managing website structure, adding and deleting pages, and organizing website content.
- **Templates and Themes:** The software provided built-in templates and themes to help users create professional-looking websites quickly.
- **Microsoft Office Integration:** Its integration with other Microsoft Office applications allowed for easy use of familiar tools for content creation.
- **Drag-and-Drop Layout:** Users could easily arrange elements on a page by simply dragging and dropping them into place.
- **Support for Advanced Features:** FrontPage could also handle more complex features like forms, discussion boards, and database integration.

Historical Significance

- **Popularity in the "Golden Era":** During the late 1990s and early 2000s, FrontPage was a popular choice for small businesses, hobbyists, and aspiring web designers
- **Bridging the Gap:** It democratized web development by making it accessible to users who were not programmers

Why it is discontinued

- FrontPage was eventually discontinued and replaced by Microsoft Expression Web and SharePoint Designer

3.5.5. MS Visual Interdev

Microsoft Visual Interdev was a discontinued, integrated development environment (IDE) for building dynamic, data-driven web applications, particularly for the Microsoft Active Server Pages (ASP) platform, offering tools for site management, database connectivity, server-side scripting, and team collaboration. Launched in the late 1990s, it was a key component of the Visual Studio

suite but was eventually superseded by the web development tools within **Visual Studio .NET** and the **ASP.NET** framework.

Key Features:

- **Integrated Development Environment:** Provided a single workspace for building and managing complex web applications.
- **Active Server Pages (ASP) Support:** Enabled developers to build server-based web applications using the ASP framework, which offered open database access, application state management, and server-side scripting.
- **Database Integration:** Included tools for connecting web applications to databases, making sites more interactive and smarter.
- **Site Management and Content Editing:** Offered tools for managing the complete web application life cycle, from content editing to publishing.
- **Team Development:** Included features for supporting teams of developers working together on web projects.
- **Scripting & Debugging:** Provided features like code completion, syntax highlighting, and an integrated debugger for VBScript and JScript.

Evolution and Legacy:

- Visual Interdev's core concepts, such as code completion, syntax highlighting, and integrated database tools, heavily influenced the modern Visual Studio interface.
- It was an important step in Microsoft's strategy to provide comprehensive tools for professional web developers, including support for building component-based applications.
- The tool's features, particularly its scripting and database integration, made it a powerful choice for enterprise-class applications built on Microsoft's BackOffice infrastructure.

3.5.6. Macromedia Dreamweaver

Macromedia Dreamweaver, later acquired by Adobe and now known as Adobe Dreamweaver, is a web design and development application used to create, code, and manage websites. It provides a comprehensive environment for web developers and designers, catering to both visual design and hand-coding.

Key aspects of Macromedia/Adobe Dreamweaver in web development include:

- **Visual Design Interface:** Dreamweaver offers a "Design View" that allows users to visually design web pages with a drag-and-drop interface, similar to a desktop publishing program. This enables users to arrange elements, insert images, and create layouts without needing to write code directly.
- **Code Editor:** It includes a robust code editor that supports various web languages like HTML, CSS, JavaScript, PHP, and more. This editor features syntax highlighting, code completion, real-time syntax checking, and code introspection to assist in writing clean and accurate code.
- **Split View:** Dreamweaver's "Split View" allows users to see both the visual design and the underlying code simultaneously, facilitating a direct connection between the visual representation and its code.
- **Site Management:** It provides tools for organizing and managing website files and folders, enabling efficient site structure and maintenance.
- **Integration with Web Technologies:** Dreamweaver supports various web technologies, including server-side scripting languages, databases, and frameworks, allowing for the creation of dynamic and interactive websites.
- **FTP/SFTP/WebDAV Support:** It integrates with protocols like FTP, SFTP, and WebDAV for uploading and managing website files on remote web servers.
- **Version Control:** Later versions of Dreamweaver also incorporated support for version control systems like Subversion (SVN) to manage changes in code collaboratively.